GL-DETR: Global-to-Local Transformers for Small Ship Detection in SAR Images

Cong Li, Yongqiang Hei, Lihu Xi, Wentao Li, *Member, IEEE*, and Zhu Xiao, *Senior Member, IEEE*

***Abstract*— Transformer-based methods have demonstrated their potential capabilities in ship detection of synthetic aperture radar (SAR) images. However, their exclusive reliance on global context information hinders the precise localization of small ships, leading to suboptimal detection performance. In this work, a global-to-local detection transformer (GL-DETR) framework is proposed to enhance the detection accuracy of small ships in SAR images. In GL-DETR, the decoder is constituted of a global layer and a carefully designed local layer. Within the global layer, object queries fully interact with global context information. Furthermore, a local interaction attention (LIA) module is designed in the local layer, with the purpose of refining and enriching the object query features through local multiscale region of interest (ROI) information. Additionally, a multiscale information enhancement (MIE) module is introduced to enhance the high-frequency information of small ships through Gaussian filtering. Numerical results demonstrate that GL-DETR outperforms baseline by 3.71% and 4.89% on classical SAR datasets LS-SSDD and HRSID, respectively. These results demonstrate the effectiveness and superiority of the proposed strategy.**

***Index Terms*— Feature enhancement, global-to-local transformer, small ship detection, synthetic aperture radar (SAR) image.**

I. Introduction

SYNTHETIC aperture radar (SAR) emerges as a high-resolution microwave imaging radar system capable of all-weather and all-time observation, which holds significant importance in both civil and industrial fields. By taking advantage of SAR, object detection in SAR images has attracted widespread attention in various domains, such as disaster relief, ocean monitoring, and city planning [1], [2]. Recently, significant progress has been achieved in large-scale object detection in SAR images [4], [5], [6]. However, small object detection is still confronted with distinct challenges, including inadequate detailed information and being easily submerged in complex backgrounds [7], [8], which extensively occurs in practical security tasks, such as sea rescue of ships and unmanned aerial vehicle (UAV) identification. Consequently, effective strategies have to be developed to address these issues.

In general, small object detection for SAR images can be mainly classified into traditional and deep learning-based methods. Traditional methods, such as constant false-alarm rate (CFAR) [9], always set a threshold to distinguish between the objects and background by statistical modeling. However, due to their dependence on manually constructed feature extractors, the detection performance is unsatisfactory when confronted with complex scenes. On the contrary, deep learning-based methods show more significant advantages due to their automated feature extraction and strong feature representation capabilities. Generally, they can be categorized into: convolutional neural network (CNN) based and transformer-based architectures.

CNN-based methods can be further divided into two-stage and one-stage ones. The two-stage methods, such as region-CNN (R-CNN) [12], Faster R-CNN [13], and Cascade R-CNN [14], typically with higher accuracy. They first extract candidate regions that may contain objects and then perform classification and regression. However, two-stage methods are confronted with the challenge of real-time detection due to the substantial time required to generate candidate regions. To overcome this shortcoming, one-stage methods, such as single shot multibox detector (SSD) [15] and you only look once (YOLO) series [16], [17], directly predict the class and location of the object from the extracted features. They can achieve a better balance between accuracy and detection speed. However, CNN-based methods easily result in high false alarm rates for small objects due to the lack of contextual information required to fully understand the scene.

In recent years, the fully end-to-end object detector based on detection transformer (DETR) [20] has been put forward. Different from CNNs that focus only on local context, DETR captures global information and large-scale dependencies through cross-attention, demonstrating its powerful capability in various computer vision tasks [21], [22]. Nevertheless, DETRs exhibit relatively poor performance in detecting small objects. To address this issue, diverse improvements have been suggested in DETR architectures. In [24], deformable DETR is introduced, which incorporates multiscale local information around reference points through deformable convolutions, thus improving the accuracy of small object detection. In [25],

Received 24 April 2024; revised 23 June 2024, 7 August 2024, and 12 September 2024; accepted 12 September 2024. Date of publication 16 September 2024; date of current version 1 October 2024. This work was supported in part by the National Natural Science Foundation under Grant 62376204, Grant 62371369, and Grant 62476208; in part by the National Natural Science Foundation for Distinguished Young Scholar under Grant 62425103; in part by Shaanxi Natural Science Foundation under Grant 2023-JC-YB-505; and in part by the Aeronautical Science Foundation of China under Grant 20240020081001. *(Corresponding author: Yongqiang Hei.)*

Cong Li, Yongqiang Hei, and Lihu Xi are with the State Key Laboratory of ISN, Xidian University, Xi'an 710071, China (e-mail: licong31415@163.com; yqhei@mail.xidian.edu.cn; 2359925459@qq.com).

Wentao Li is with the Science and Technology on Antenna and Microwave Laboratory, Department of Electronic Engineering, Xidian University, Xi'an 710071, China (e-mail: wtli@mail.xidian.edu.cn).

Zhu Xiao is with the College of Computer Science and Electronic Engineering, Hunan University, Changsha 410082, China (e-mail: zhxiao@hnu.edu.cn).

Digital Object Identifier 10.1109/LGRS.2024.3461212

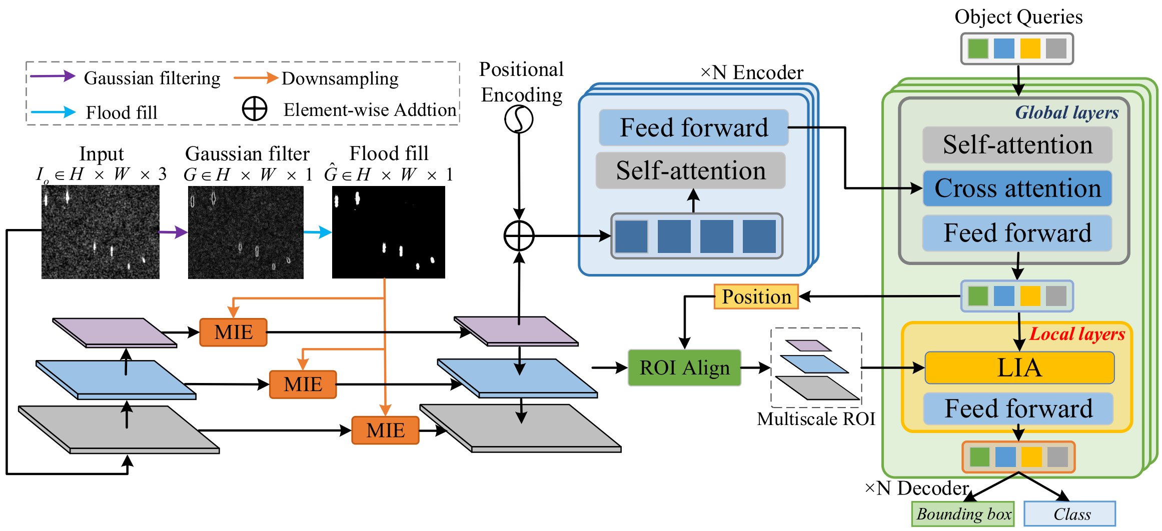


Fig. 1. Overall framework of the GL-DETR. GL-DETR retains the main encoder-decoder architecture of DETR while introducing the MIE module and GL decoder.

noisy ground truth edges are employed as training samples, helping the model learn to reconstruct the original edges, which leads to a remarkable improvement in accuracy. In [26], the robustness of the model is enhanced by using a late decoder to directly process intermediate queries from the predecoder.

In contrast to previous research, this work proposes a global-to-local DETR (GL-DETR) framework, which develops a coarse-to-fine strategy for precise localization of the bounding box. In GL-DETR, the decoder consists of a global layer and a carefully designed local layer. The global layer is to enable object queries to perceive global features, thereby obtaining a coarse localization of the bounding box. Furthermore, in the local layer, a local interaction attention (LIA) module is introduced to enrich the object query features through local multiscale region of interest (ROI) information. This module facilitates the acquisition of local and spatial information about the ship, thus further refining the localization of the bounding boxes. Additionally, due to the information of small ships is eroded as the neural network deepens, a multiscale information enhanced module is proposed to enhance the representation of small ships within multiscale features.

Briefly, the main contributions of this letter are summarized as follows:

1) To the best of our knowledge, this is the first work to explore the transformer to detect the SAR small ship through a coarse-to-fine strategy, which can fully enrich the features by fusing global and local information.
2) Two efficient components, multiscale information enhancement (MIE) and LIA, are developed to enhance the feature representation of small ships and refine the localization of the bounding box, respectively.
3) Extensive experiments on the challenging LS-SSDD and HRSID datasets show that the proposed GL-DETR exhibits superior performance.

II. Methods

A. Overview Architecture

Fig. 1 shows an overview architecture of the global-local DETR, in which the regression of bounding boxes is divided into two steps. The connection between the encoder and decoder is retained, which effectively facilitates the interaction between object query embeddings and the encoder outputs. The first step involves a rough localization based on the global context information. In the second step, LIA embeds the high-frequency information of small ships into features, and then the bounding boxes are further refined by the local ROI information. In this way, it enables the model to realize precise ship localization. Besides, position coding is introduced to enhance object detection by embedding spatial context and improving discrimination in complex environments.

B. Multiscale Information Enhanced

As we know, the features of small objects tend to erode as the network deepens. Consequently, a MIE module is proposed to enhance the feature representation of small ships. As shown in Fig. 2, a Gaussian filter filling (GFF) algorithm is introduced to extract high-frequency information from the images. In GFF, ship contours are first extracted from the original image I_0 using the Gaussian filter. These contours are then filled using a flood-fill algorithm [23]. Finally, the high-frequency features pyramid is expressed as

$$\hat{G}_i = \mathrm{down}_i(\mathrm{full}(g(I_0))) \tag{1}$$

where $g(\cdot)$ represents the convolution operator with Gaussian filter, $\mathrm{full}(\cdot)$ denotes the flood fill algorithm and $\mathrm{down}(\cdot)$ is the twofold downsampling. In essence, these high-frequency features contain key details at various scales. Subsequently, in the ith layer, the feature F_i is multiplied by the high-frequency feature $\hat{G}_i$, and then further multiplied with the attention mask. The step aims to guide the attention to key regions while suppressing background noise and redundant information. The features F_i^a are calculated as follows:

$$A_i = \mathrm{Conv}(F_i \otimes \hat{G}_i) \tag{2}$$

$$F_i^a = \left(F_i \otimes \hat{G}_i\right) \otimes A_i \oplus F_i \tag{3}$$

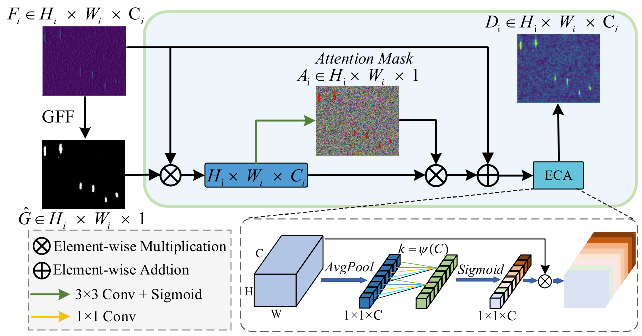


Fig. 2. Detailed illustration of MIE, which enhances the feature representation of small ships by utilizing high-frequency information extracted from Gaussian filtering.

where $\otimes$ represents element-wise multiplication, $\oplus$ denotes element-wise addition, and $\mathrm{Conv}(\cdot)$ refers to the size of 1×1 convolution. As shown in Fig. 2, the attention mask A_i displays significantly elevated values at pixels located on ships. Furthermore, an effective channel attention (ECA) [27] module for recalibration is introduced to capture feature correlations between boundary and background regions. Specifically, global average pooling and 1-D convolution are performed according to the size of the convolution kernel expressed as follows:

$$k = \psi(C) = \left| \frac{\log_2(C)}{\gamma} + \frac{b}{\gamma} \right|_{\mathrm{odd}} \tag{4}$$

where $|t|_{\mathrm{odd}}$ denotes the nearest odd number of t, γ, and b are set to 2 and 1, respectively. The weights of each channel are then computed to finally produce the fine features as

$$D_i = \sigma\left(\mathrm{Conv1D}\left(\mathrm{AvgPool}\left(F_i^a\right)\right)\right) \cdot F_i^a \tag{5}$$

where $\mathrm{AvgPool}(\cdot)$ denotes channel-wise average pooling, $\mathrm{Conv1D}(\cdot)$ is the 1-D convolution, and σ is the sigmoid.

C. Local Interaction Attention

In the original decoder of DETR, the learnable object queries are interacted with global information and then mapped to categories and locations of ships. However, the localization of small ships remains coarse due to the lack of detailed information. To facilitate a more fine-grained regression of the bounding box, LIA is proposed by introducing multiscale ROI information. As shown in Fig. 3, given an object query feature $q_i \in \mathbb{R}^{1\times d}$, the position information is predicted by q_i, and then it is mapped into multiscale ROI $f_i^{\mathrm{ROI}} \in \mathbb{R}^{h\times w\times c}$. Meanwhile, the object query q_i is processed by the fully connected layer and expanded to $q_i^e \in \mathbb{R}^{h\times w\times c}$. Subsequently, a deep interaction is conducted between the object query $q_i^e \in \mathbb{R}^{h\times w\times c}$ and multiscale ROI $f_i^{\mathrm{ROI}} \in \mathbb{R}^{h\times w\times c}$ via cross-attention as follows:

$$\begin{aligned} Q_i &= \mathrm{Re}\left(f_i^{\mathrm{ROI}}\right) W_i^Q, \quad K_i = \mathrm{Re}\left(q_i^e\right) W_i^K \\ V_i &= \mathrm{Re}\left(q_i^e\right) W_i^V \end{aligned} \tag{6}$$

where $\mathrm{Re}(\cdot)$ reshapes the features to $\mathbb{R}^{hw\times c}$ by combining the height (h) and width (w) dimensions into a single dimension,

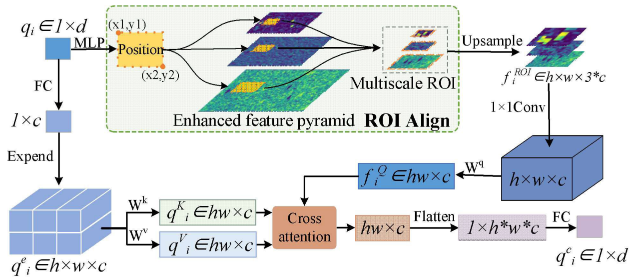


Fig. 3. Detailed illustration of LIA, which utilizes the local detail information from the ROI to interact with object queries, resulting in more accurate localization of small ships.

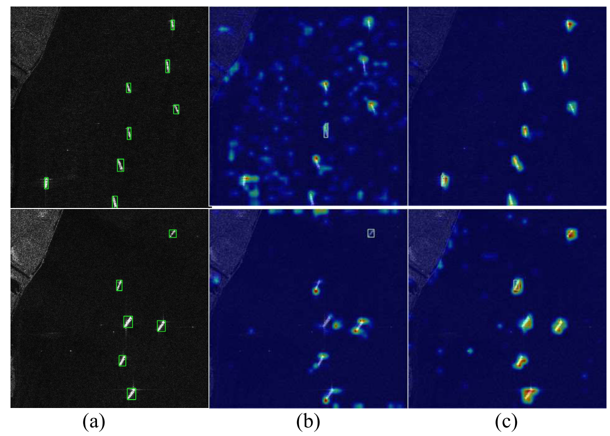


Fig. 4. Comparison of feature heatmaps between MIE and FPN. (a) Ground truth. (b) FPN. (c) MIC.

and $W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{c\times c}$ denote three weight matrices. The attention vectors are calculated by

$$f_i^{\mathrm{CA}} = \mathrm{softmax}\left(\frac{Q_i K_i^{\mathrm{T}}}{\sqrt{d_k}}\right) V_i \tag{7}$$

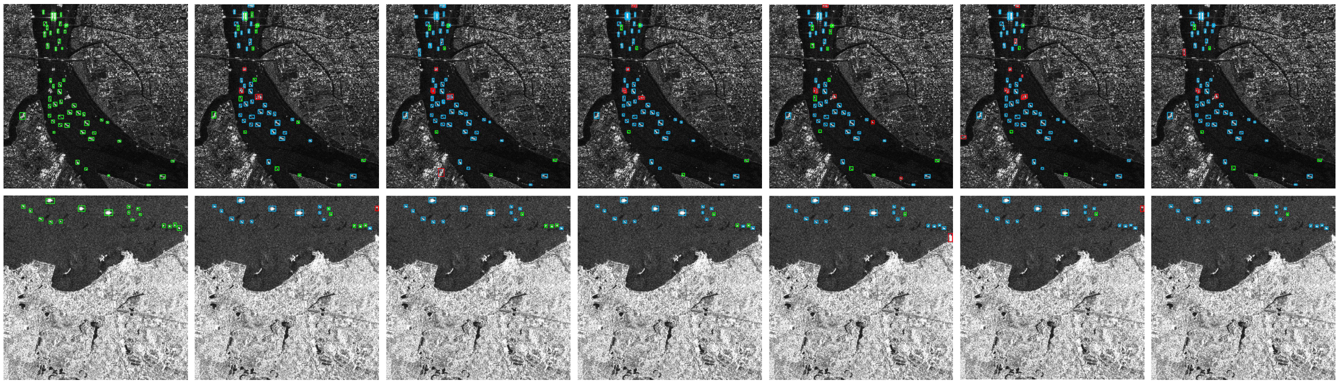
where d_k is the dimension of c, K_i^{T} refers to the transpose of K_i. Finally, $f_i^{\mathrm{CA}} \in \mathbb{R}^{hw\times c}$ is flattened to $\mathbb{R}^{1\times hwc}$ by concatenating along the last dimension, and further transformed into $q_i^c \in \mathbb{R}^{1\times d}$ through a fully connected layer.

III. Experiments

A. Implementation Details

All modules are trained for 200 iterations on PyTorch 1.8.1 using NVIDIA RTX 3090 GPUs. The initial learning rate is set to 0.001 and follows a cosine annealing schedule. The backbone network is pretrained CSP-DarkNet53. Two publicly mainstream SAR datasets are utilized for validation as follows:

1) *LS-SSDD-V1.0 [10]:* The dataset is taken by Sentinel-1 satellites and released in 2017, it contains a total of 6015 labeled ships, with 96.2% of them smaller than 32×32 pixels.
2) *HRSID [3]:* HRSID is released in 2020 and obtained from the TerraSAR-X satellites. The dataset contains a total of 16 951 labeled ships, 50.3% of which are small ships.



(a) (b) (c) (d) (e) (f) (g)

Fig. 5. Comparison of detection results from various methods. The first line is LS-SSDD-v1.0, and the second line is HRSID. Note that the blue boxes, the green boxes, and the red boxes are detected ships, missed ships, and false alarm ships, respectively. (a) Ground Truth. (b) Faster R-CNN. (c) DETR. (d) DINO. (e) Focus-DETR. (f) Deformable-DETR. (g) GL-DETR (ours).

B. Evaluation Metrics

The precision, recall, $F1$-score, and average precision are employed as evaluation metrics, which are defined as follows:

$$\text{Precision} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}}} \tag{8}$$

$$\text{Recall} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}} \tag{9}$$

$$F1\text{-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{10}$$

$$\text{AP} = \int_0^1 P(R)dR \tag{11}$$

where N_{TP}, N_{FP}, and N_{FN} denote the number of correctly detected ships, false alarms, and missing ships, respectively. $P(R)$ represents the precision when the recall is R. AP_{50} and AP_{75} correspond to AP calculated at IoU thresholds of 0.5 and 0.75, respectively.

C. Ablation Studies

To validate the effectiveness of MIE and LIA, ablation studies are conducted using Deformable-DETR [24] as the baseline. As shown in Table I, MIE contributes baseline gain by 1.88% and 2.48% improvement of AP_{50} on LS-SSDD-V1.0 and HRSID, respectively. And LIA also improves baseline by 2.56% and 3.90% AP_{50} on LS-SSDD-V1.0 and HRSID, respectively. This improvement is mainly attributed to MIE effectively enhancing the feature representation of small ships, and the detailed information of ROI is utilized fully by LIA for careful localization. Furthermore, as shown in Fig. 4, the features processed by MIE significantly improve the attention on small ships compared to the original features and the features via the feature pyramid network (FPN). Moreover, compared with the baseline, GL-DETR improves AP_{50} by 7.52% and 6.55% on LS-SSDD-V1.0 and HRSID, respectively.

TABLE I

ABLATION STUDIES OF GL-DETR ON LS-SSDD-V1.0, THE BASELINE IS DEFORMABLE-DETR

MIE	LIA	LS-SSDD			HRSID		
		F1-score	AP_{50}	AP_{75}	F1-score	AP_{50}	AP_{75}
-	-	76.38	71.47	12.13	86.11	87.24	78.73
✓	-	77.51	73.35	13.48	87.45	89.72	80.18
-	✓	78.06	74.03	14.19	88.89	91.14	91.34
✓	✓	**78.71**	**75.18**	**14.96**	**89.47**	**92.13**	**82.17**

TABLE II

COMPARISON RESULTS WITH EXISTING ADVANCED DETECTORS ON LS-SSDD-V1.0

Methods	Precision	Recall	F1-score	AP_{50}	AP_{75}	FPS
Faster R-CNN	76.27	78.21	77.23	71.64	11.56	24.5
SARNas	77.36	76.49	76.92	72.56	12.87	16.4
DETR	73.16	74.28	73.72	69.53	10.54	10.4
Focus-DETR	76.91	76.17	76.54	73.04	13.16	**19.2**
DINO	76.77	76.18	76.47	72.86	13.18	13.5
Deformable-DETR	75.27	77.53	76.38	71.47	12.13	16.2
GL-DETR	**78.13**	**79.29**	**78.71**	**75.18**	**14.96**	18.5

D. Comparison With Advanced Methods

To verify the effectiveness of the proposed method, comparisons are implemented with existing advanced methods, such as Faster R-CNN, SARNas [18], DETR, Focus-DETR [11], DINO [19], and Deformable-DETR. As shown in Table II, GL-DETR outperforms all the other models on AP_{50} value on the LS-SSDD-V1.0 dataset. Specifically, it outperforms DETR by 5.65% and Deformable-DETR by 3.71%. This result validates the superiority of our proposed method. On the other hand, as shown in Table III, GL-DETR consistently shows at the highest level on the HRSID with AP_{50} of 92.13%, which is 12.45% higher than DETR and 5.02% higher than DINO. In addition, GL-DETR achieves the highest $F1$-score. This achievement is attributed to the significantly enhanced feature representation of small ships by MIC and the global-local design in the encoder, which allows for fine-grained localization of small ships.

TABLE III
COMPARISON RESULTS WITH EXISTING ADVANCED DETECTORS ON HRSID

Methods	Precision	Recall	F1-score	AP_{50}	AP_{75}	FPS
Faster R-CNN	86.48	76.91	81.41	82.13	71.86	**23.1**
SARNas	89.26	**88.35**	88.80	90.82	78.23	14.9
DETR	84.86	83.20	84.02	79.68	72.47	8.4
Focus-DETR	89.59	85.69	87.60	88.96	79.89	17.7
DINO	88.67	84.78	86.68	87.22	79.67	12.6
Deformable-DETR	88.61	83.75	86.11	87.24	78.73	14.9
GL-DETR	**91.45**	87.57	**89.47**	**92.13**	**82.17**	16.1

E. Visual Comparative Analysis

In this section, comparative visualization results are conducted with existing models, including Faster R-CNN, DETR, Focus-DETR, DINO, and DN-DETR on LS-SSDD-V1.0 and HRSID, respectively. The corresponding results of these models are presented in detail in Fig. 5. It can be seen that GL-DETR can locate and identify small ships in complex scenes more accurately than the other models, significantly reducing false alarms and missed alarms, especially in scenes with densely arranged boats. These visual pieces of evidence show that GL-DETR exhibits significant advantages in the challenge of small ship detection in SAR images.

IV. CONCLUSION

In this letter, an effective end-to-end transformer framework called GL-DETR has been proposed to enhance the detection accuracy of small ships in SAR images. In GL-DETR, a global-to-local decoder architecture has been designed, which can utilize global context information and local detail information to achieve coarse-to-fine localization. Specifically, the global layer aims for coarse localization, and in the local layer, the LIA module has been introduced to refine the localization of the bounding box by interacting with local multiscale ROI information. In addition, the MIE module has been proposed to enhance the feature representation of small ships. Numerical experiments have been carried out to verify the effectiveness of the proposed GL-DETR in small-ship detection.

REFERENCES

[1] J. Cheng, F. Zhang, D. Xiang, Q. Yin, and Y. Zhou, "PolSAR image classification with multiscale superpixel-based graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5209314.

[2] H. Chang, X. Fu, J. Dong, J. Liu, and Z. Zhou, "MLSDNet: Multiclass lightweight SAR detection network based on adaptive scale distribution attention," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[3] S. Wei, X. Zeng, Q. Qu, M. Wang, H. Su, and J. Shi, "HRSID: A high-resolution SAR images dataset for ship detection and instance segmentation," *IEEE Access*, vol. 8, pp. 120234–120254, 2020.

[4] T. Zhang and X. Zhang, "A mask attention interaction and scale enhancement network for SAR ship instance segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[5] P. Guo, T. Celik, N. Liu, and H.-C. Li, "Break through the border restriction of horizontal bounding box for arbitrary-oriented ship detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[6] S. Gao, J. M. Liu, Y. H. Miao, and Z. J. He, "A high-effective implementation of ship detector for SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[7] M. Zha, W. Qian, W. Yang, and Y. Xu, "Multifeature transformation and fusion-based ship detection with small targets and complex backgrounds," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[8] L. Bai, C. Yao, Z. Ye, D. Xue, X. Lin, and M. Hui, "A novel anchor-free detector using global context-guide feature balance pyramid and united attention for SAR ship detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[9] X. Qin, S. Zhou, H. Zou, and G. Gao, "A CFAR detection algorithm for generalized gamma distributed background in high-resolution SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 4, pp. 806–810, Jul. 2013.

[10] T. Zhang et al., "LS-SSDD-v1.0: A deep learning dataset dedicated to small ship detection from large-scale Sentinel-1 SAR images," *Remote Sens.*, vol. 12, no. 18, p. 2997, Sep. 2020.

[11] D. Zheng, W. Dong, H. Hu, X. Chen, and Y. Wang, "Less is more: Focus attention for efficient DETR," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 6674–6683.

[12] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2014, pp. 580–587.

[13] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[14] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[15] W. Liu et al., "SSD: Single shot MultiBox detector," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2016, pp. 21–37.

[16] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[17] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Apr. 2018, pp. 1–6.

[18] W. Du et al., "SARNas: A hardware-aware SAR target detection algorithm via multiobjective neural architecture search," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5212923.

[19] H. Zhang et al., "DINO: DETR with improved DeNoising anchor boxes for end-to-end object detection," 2022, *arXiv:2203.03605*.

[20] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 213–229.

[21] S. d'Ascoli, H. Touvron, M. Leavitt, A. Morcos, G. Biroli, and L. Sagun, "ConViT: Improving vision transformers with soft convolutional inductive biases," 2021, *arXiv:2103.10697*.

[22] X. Cao, P. Yuan, B. Feng, and K. Niu, "CF-DETR: Coarse-to-fine transformers for end-to-end object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 185–193.

[23] Y. He, T. Hu, and D. Zeng, "Scan-flood Fill(SCAFF): An efficient automatic precise region filling algorithm for complicated regions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2019, pp. 761–769.

[24] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2020, pp. 1–16.

[25] F. Li, H. Zhang, S. Liu, J. Guo, L. M. Ni, and L. Zhang, "DN-DETR: Accelerate DETR training by introducing query denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13609–13617.

[26] F. Chen, H. Zhang, K. Hu, Y.-K. Huang, C. Zhu, and M. Savvides, "Enhanced training of query-based object detection via selective query recollection," 2022, *arXiv:2212.07593*.

[27] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-net: Efficient channel attention for deep convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11531–11539.